

# PHY 211

## Lecture 4 Notes

By

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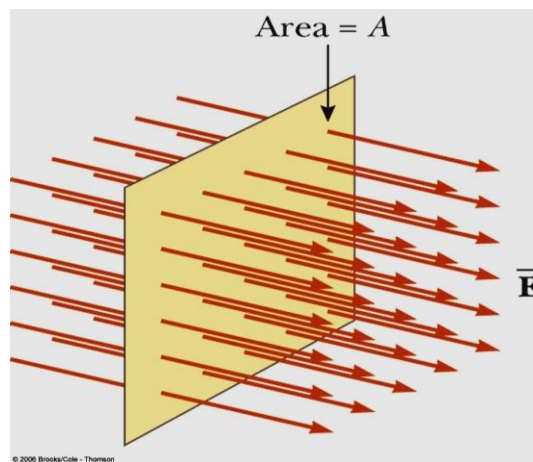
# Chapter 2

## Electric Flux & Gauss's Law

### ➤ The electric flux:

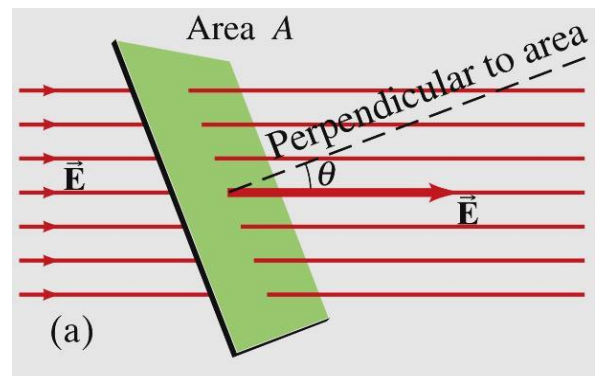
- The electric flux is the number of electric field lines penetrating some surface area perpendicular to it.
- Electric flux is the product of the magnitude of the electric field and the surface area  $A$  perpendicular to the field.

$$\phi_E = EA \left( \frac{\text{N} \cdot \text{m}^2}{\text{C}} \right)$$



- The field lines may make some angle  $\theta$  with the perpendicular to the surface then:

$$\phi_E = \vec{E} \cdot \vec{A} = EA \cos \theta \left( \frac{\text{N} \cdot \text{m}^2}{\text{C}} \right)$$



- The flux is a maximum when the surface is perpendicular to the field (i.e.  $\theta = 0, \cos 0 = 1$ ) ( $(\phi)_{max} = EA$ )
- The flux is a minimum “Zero” when the surface is parallel to the field (i.e.  $\theta = 90, \cos 0 = 0$ ) ( $(\phi)_{min} = 0$ )

- **Example 1:** A rectangular surface (0.16 m & 0.38 m) is oriented in a uniform electric field of 580 N/C.
  - a) What is the maximum possible electric flux through the surface and?
  - b) What is the electric flux through if the surface is making 30° with the field?

**Solution**

$$a) (\phi)_{max} = EA = 580 \times (0.16 \times 0.38) = 35.364 \frac{\text{N} \cdot \text{m}^2}{\text{C}}$$

$$b) \phi = EA \cos \theta = 580 \times (0.16 \times 0.38) \cos 60 = 17.682 \frac{\text{N} \cdot \text{m}^2}{\text{C}}$$


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- **Gauss' Law:**

The net electric flux  $\phi_E$  through any closed surface is equal to the net charge  $q_{in}$  inside the surface divided space permittivity  $\epsilon_0$  and is independent of the shape of the surface:

**Gauss's law:**      $\phi = \oint \vec{E} \cdot d\vec{A} = \oint E dA \cos \theta = \frac{q_{in}}{\epsilon_0}$

→ closed surface integration

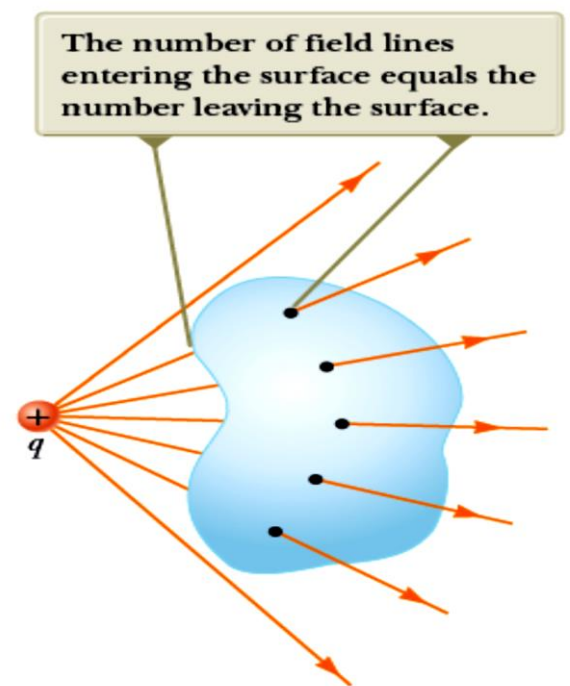
$q_{in}$ : net charge inside the surface      $\epsilon_0$ : permittivity of free space

$\theta$ : angle between electric field and the normal to the surface

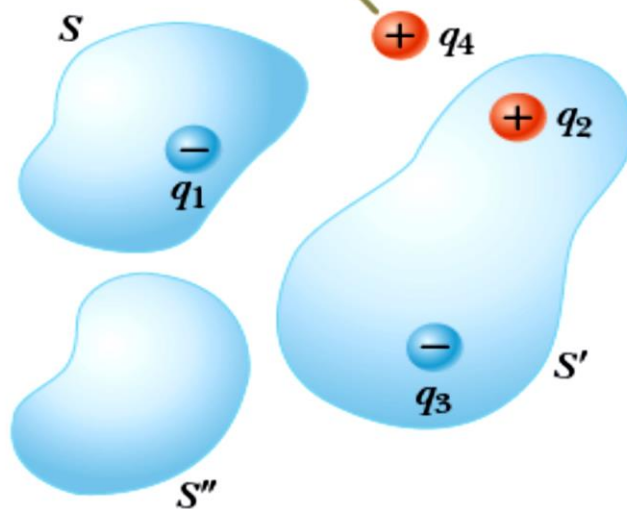
- For a point charge located outside a closed surface, any electric field line entering the surface leaves the surface at another point. The number of electric field lines entering the surface equals the number leaving the surface.

**(outward flux +ve, inward flux -ve)**

- Therefore, the net electric flux through a closed surface that surrounds no charge is zero.



Charge  $q_4$  does not contribute to the flux through any surface because it is outside all surfaces.



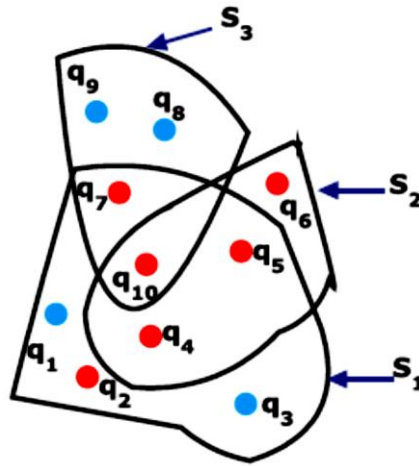
$$\phi_S = \frac{q_1}{\epsilon_0}$$

$$\phi_{S'} = \frac{q_2 + q_3}{\epsilon_0}$$

$$\text{If } q_2 = -q_3 \text{ then } \phi_{S'} = 0$$

$$\phi_{S''} = 0$$

- **Problem:** In the shown figure, calculate the electric flux through surfaces S1, S2 and S3 if  $q_1 = q_3 = q_8 = q_9 = +2 \mu\text{C}$  and  $q_2 = q_4 = q_5 = q_6 = q_7 = q_{10} = -2 \mu\text{C}$ .



### Solution

$$\phi_{S1} = q_1 + q_2 + q_3 + q_4 + q_5 + q_7 + q_{10}$$

$$\phi_{S1} = \frac{(2 - 2 + 2 - 2 - 2 - 2 - 2) \times 10^{-6}}{\epsilon_0} = \frac{-6 \times 10^{-6} \text{ N} \cdot \text{m}^2}{\epsilon_0 \text{ C}} \text{ (inward)}$$

$$\phi_{S2} = q_4 + q_5 + q_6 + q_{10}$$

$$\phi_{S2} = \frac{(-2 - 2 - 2 - 2) \times 10^{-6}}{\epsilon_0} = \frac{-8 \times 10^{-6} \text{ N} \cdot \text{m}^2}{\epsilon_0 \text{ C}} \text{ (inward)}$$

$$\phi_{S3} = q_7 + q_8 + q_9 + q_{10}$$

$$\phi_{S3} = \frac{(-2 + 2 + 2 - 2) \times 10^{-6}}{\epsilon_0} = 0$$

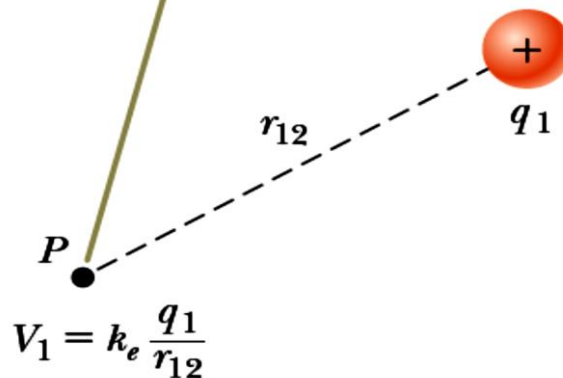
# Chapter 3

## Electric Potential, potential difference and potential energy

- **Electric potential:** It is the work done to bring a unit positive charge from infinity to a certain point.
- The electric potential is a scalar quantity, +ve for +ve charge and –ve for –ve charge

$$V = \frac{KQ}{r} \quad \left( \text{unit: Volt or } \frac{\text{J}}{\text{C}} \right)$$

A potential  $k_e q_1 / r_{12}$  exists at point  $P$  due to charge  $q_1$ .



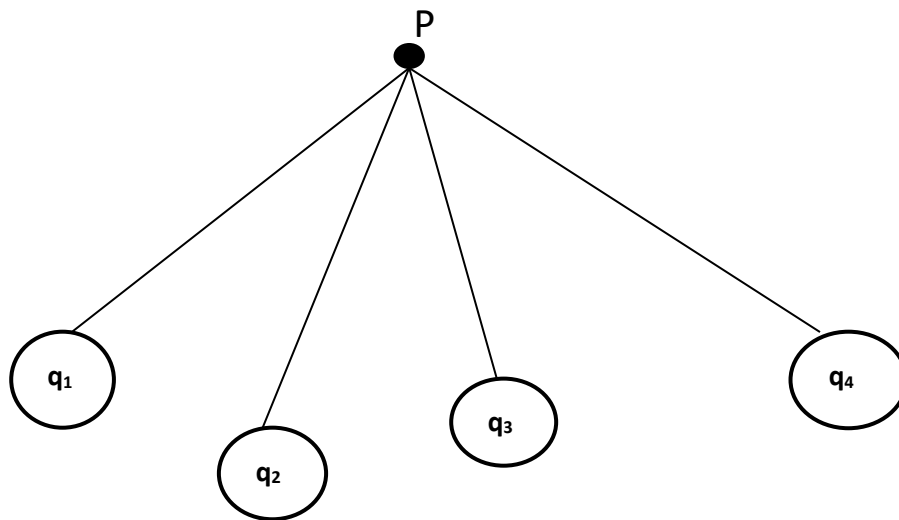
- Electric potential due to several charges:**

The net potential at a point due to a group of point charges ( $q_1, q_2, q_3, \dots, q_n$ ) is given by:

$$V_{\text{net}} = V_1 + V_2 + V_3 + \dots + V_n$$

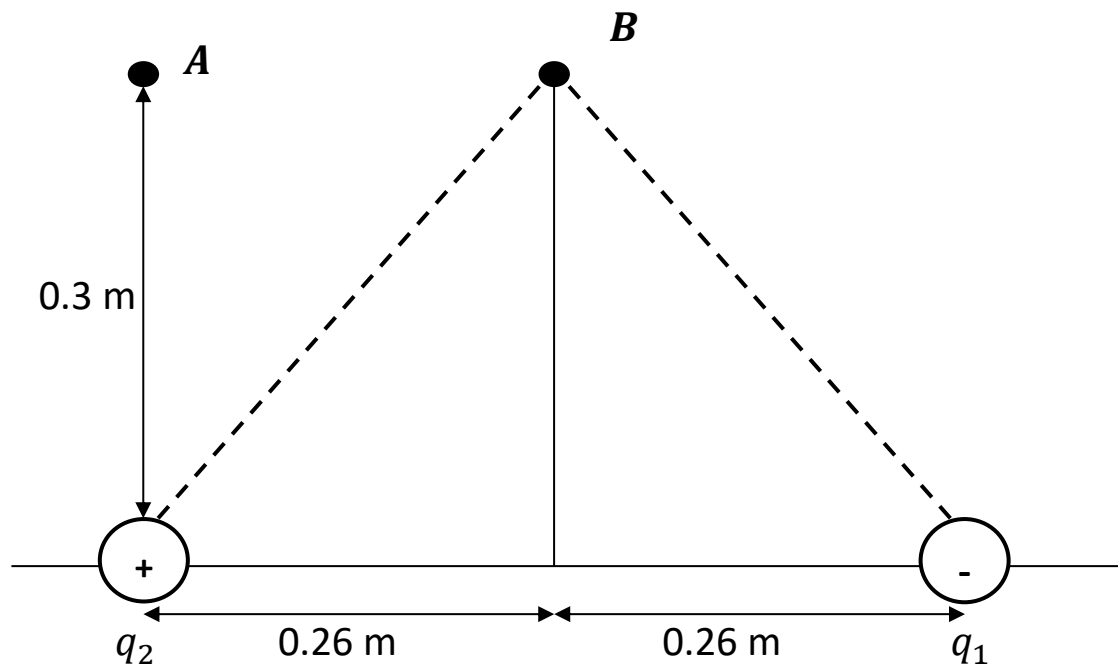
$$V_{\text{net}} = \sum_{i=1}^n V_i = K \sum_{i=1}^n \frac{q_i}{r_i} =$$

$$V_{\text{net}} = K \left[ \frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots + \frac{q_n}{r_n} \right]$$



$$V_P = K \left[ \frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \frac{q_4}{r_4} \right]$$

- **Example 1:** Calculate the electric potential at points A & B due to charges  $q_1 = -50 \mu\text{C}$  & and  $q_2 = 50 \mu\text{C}$  as shown in figure.



### Solution

At point A:

$$V_A = V_{A1} + V_{A2} = \frac{Kq_1}{r_{A1}} + \frac{Kq_2}{r_{A2}} = K \left[ \frac{q_1}{r_{A1}} + \frac{q_2}{r_{A2}} \right]$$

$$V_A = 9 \times 10^9 \left[ \frac{(-50 \times 10^{-6})}{0.6} + \frac{(50 \times 10^{-6})}{0.3} \right] = 7.5 \times 10^5 \text{ V}$$

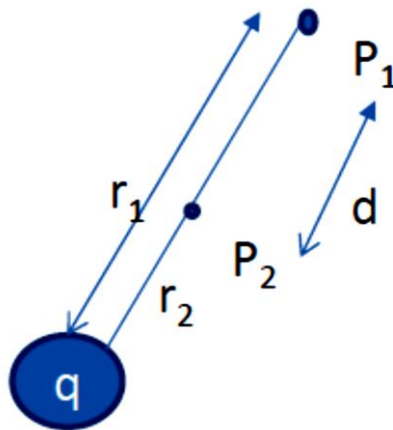
At point B:

$$V_B = V_{B1} + V_{B2} = \frac{Kq_1}{r_{B1}} + \frac{Kq_2}{r_{B2}} = K \left[ \frac{q_1}{r_{B1}} + \frac{q_2}{r_{B2}} \right]$$

$$V_A = 9 \times 10^9 \left[ \frac{(-50 \times 10^{-6})}{0.4} + \frac{(50 \times 10^{-6})}{0.4} \right] = 0 \text{ V}$$

➤ **Potential difference:** It is the work done to transfer a unit positive charge between two points

$$\Delta V = V_{P_2} - V_{P_1} = \frac{Kq}{r_2} - \frac{Kq}{r_1} \quad (\text{unit: Volt})$$



➤ **Potential energy**

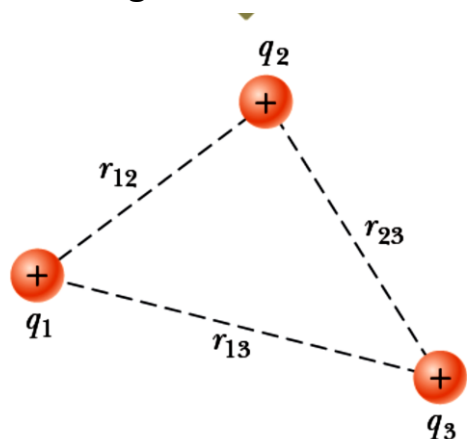
- The electric potential energy of a pair of charges is given by:

$$U = K \frac{q_1 q_2}{r} \quad (\text{unit: Joule})$$

- The total potential energy of the system of three charges

$$U = U_{12} + U_{13} + U_{23}$$

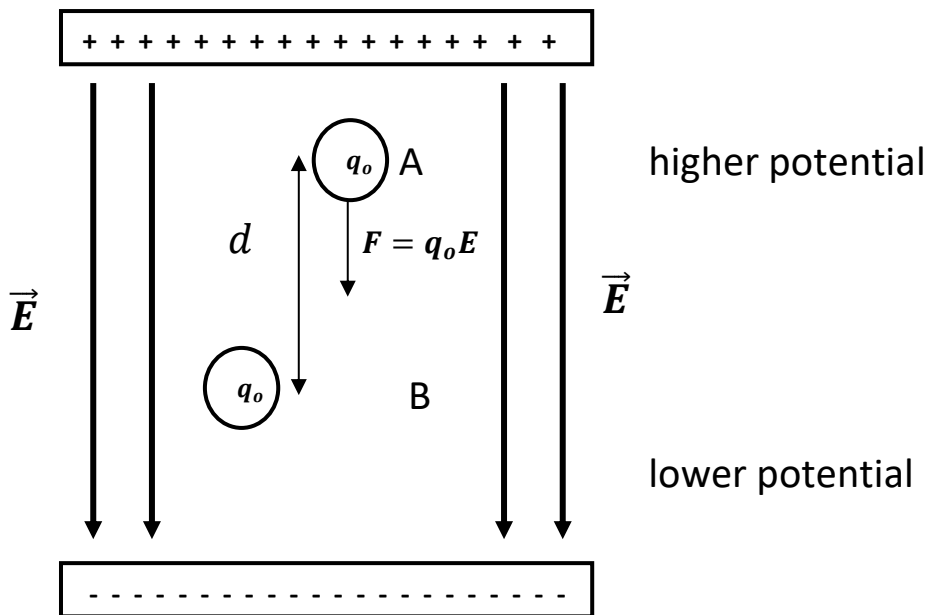
$$U = K \left[ \frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$





## ➤ The potential difference in uniform electric field:

- If a positive test charge  $q_o$  is placed in uniform electric field at point "A". It experiences to an electric force  $F = q_o E$  which displaces it to point "B".



- Electric field lines are directed from higher potential to lower potential

$$\Delta U = U_B - U_A (-ve) \quad \& \quad \Delta V = V_B - V_A (-ve) (V_B < V_A)$$

$$\Delta V = -Ed \rightarrow E = -\frac{\Delta V}{d}$$

- The electric field equals to negative the gradient of electric potential.

$$\Delta U = q_o \Delta V = -q_o E d$$

- $\Delta U$  is negative, the electric potential energy of the system decreases when the charge moves in the direction of the field.
- The work done by the electric force (field) is +ve & given by

$$\therefore W = -\Delta U = -q_o \Delta V = q_o E d = F d \quad (\text{Joule})$$

- **Example 2:**

The work done by the electric force as a test charge ( $q_o = +2 \times 10^{-6} \text{ C}$ ) moves from A to B is  $W_{AB} = +5 \times 10^{-5} \text{ J}$ .

- a) Find the difference  $\Delta U = U_B - U_A$  in the electric potential between these points.
- b) Determine the electric potential difference  $\Delta V = V_B - V_A$  between these points.

**Solution**

a)  $\Delta U = -W_{AB} = -5 \times 10^{-5} \text{ J}$

b)  $\Delta V = \frac{\Delta U}{q_o} = \frac{-5 \times 10^{-5}}{2 \times 10^{-6}} = -25 \text{ V}$

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- **Example 3:**

Find the change in electric potential energy  $\Delta U$  as a charge of (a)  $+2.2 \mu\text{C}$  or (b)  $-1.1 \mu\text{C}$  moves from a point A to a point B, given that the change in electric potential (the potential difference) between the two points is  $\Delta V = V_B - V_A = +24 \text{ V}$  (+ve & therefore point B is at higher potential compared to A).

**Solution**

a)  $\Delta U = q \cdot \Delta V = (2.2 \times 10^{-6})(24) = 5.28 \times 10^{-5} \text{ J}$

b)  $\Delta U = q \cdot \Delta V = (-1.1 \times 10^{-6})(24) = -2.64 \times 10^{-5} \text{ J}$

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- **Example 4:**

An electron in the tube of a television set is accelerated from rest through a potential difference of 5000 V.

- a) What is the change in potential energy of the electron?
- b) What is the speed of the electron ( $m = 9.1 \times 10^{-31} \text{ Kg}$ ) as a result of this acceleration?
- c) Repeat for a proton ( $m = 1.67 \times 10^{-27} \text{ Kg}$ ) that accelerates through a potential difference of 5000 V.

### Solution

- a)  $\Delta U = q \cdot \Delta V = (-1.6 \times 10^{-19})(5000) = -8 \times 10^{-16} \text{ J}$   
 (-ve sign means that when the electron moves from low to high potential loses potential energy)
- b) The potential energy lost by the electron is converted into kinetic energy, therefore

$$(\text{change in K.E}) \Delta K + (\text{change in P.E}) \Delta U = 0$$

$$\therefore \Delta K = -\Delta U$$

$$\therefore K_2 - K_1 = -\Delta U$$

Final K.E

Initial K.E

$$\frac{1}{2}mv^2 - 0 = -\Delta U$$

$$\therefore v^2 = -\frac{2\Delta U}{m_e} = -\frac{2 \times (-8 \times 10^{-16})}{9.1 \times 10^{-31}} = 1.76 \times 10^{15}$$

$$v = 4.19 \times 10^6 \text{ m/sec}$$

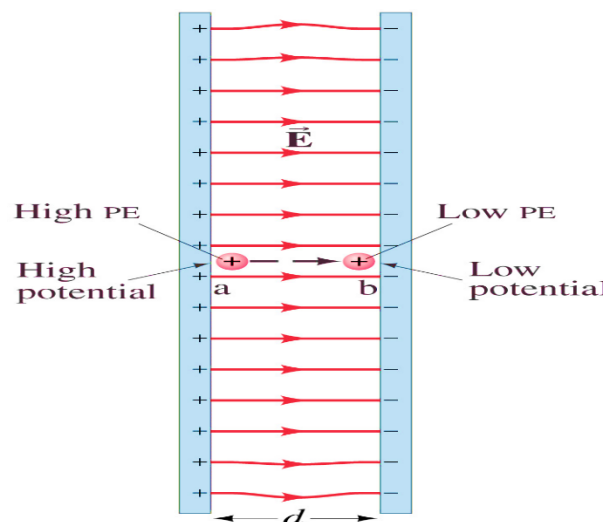
- c)  $\Delta V = -5000\text{V}$  because the protons moves from the +ve terminal to the -ve terminal

$$q_p = -q_e = +1.6 \times 10^{-19}\text{C}$$

$\Delta U = q \cdot \Delta V = (1.6 \times 10^{-19})(-5000) = -8 \times 10^{-16} \text{ J}$  (-ve sign means that when the proton moves from high potential to low potential it loses potential energy)

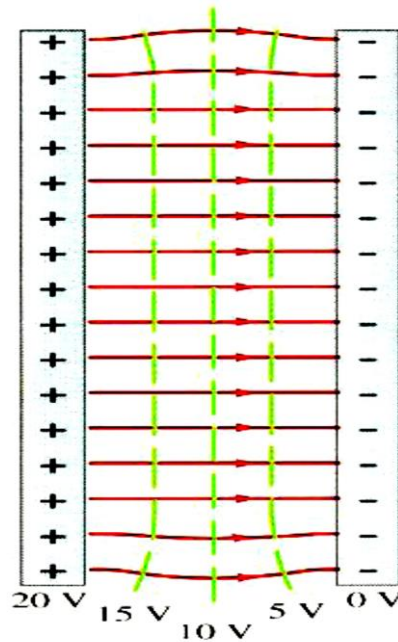
$$\begin{aligned} \therefore v^2 &= -\frac{2\Delta U}{m_p} \\ &= -\frac{2 \times (-8 \times 10^{-16})}{1.67 \times 10^{-27}} \\ &= 9.58 \times 10^{11} \end{aligned}$$

$$v = 9.79 \times 10^5 \text{ m/sec}$$



## ➤ Equipotential Lines and surfaces

An equipotential line/surface is one in which all points on it are at the same potential. That is to say, the potential difference ( $\Delta V$ ) between any two points on an equipotential line/surface is zero.



### ● Properties of equipotential line:

1. Electric field lines point from high to low potential.
2. Electric field lines are perpendicular to equipotential line.
3. Electric field lines have no component along equipotential line.
4. No work is done to move along equipotential line.